

Vansh Kapoor

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in vansh28kapoor

Education

Carnegie Mellon University (CMU)

Pittsburgh, PA

Master of Science in Machine Learning | GPA: **4.18/4**

Dec'25

- Key Courses: Deep RL (A+), Generative AI (A+), Statistics (A+), ML Optimization (A+), Adv ML (A+)
- Teaching Assistant for Deep Reinforcement Learning; delivered recitations, and mentored 8+ team projects

Indian Institute of Technology Bombay (IITB)

Mumbai, India

Bachelor of Technology in Electrical Engineering with Honors

May'24

GPA: **9.76/10** (Department Rank 6th amongst 200+ students)

Honors GPA: **10/10**

- Received Undergraduate Research Award for outstanding research conducted in Partially Observable MDPs
- Awarded AP Grade (Course Topper among 200+) in Reinforcement Learning, Adv ML, Adv Deep Learning

Publications

- V. Kapoor, J. Nair. "MDPs with State Sensing Costs." *Accepted at AISTATS'26* (Paper)
- V. Kapoor, A. Gupta, H. Chen, A. Beniwal, J. Huang, A. Kumar. "TRIM: Hybrid Inference via Targeted Stepwise Routing in Multi-Step Reasoning Tasks" *Accepted at ICLR'26* (Paper)

Industry & Research Experience

Amazon | Applied Science Intern (ICLR 2026) | San Francisco, CA

May'25 - Aug'25

- Developed RL-based **dynamic-LLM routing** for reasoning tasks (math, code generation), surpassing **BERT** and Similarity-Weighted ranking, achieving strong-model performance with **80% fewer expensive tokens** on AIME.

Google DeepMind | Research Collaborator | Bangalore, India

Jul'23 - Aug'24

- Designed look-ahead algorithms for **early rumor detection** on social graphs and adapted them for large-scale networks to nullify **bot attacks** and **adversarial misinformation propagation** on content platforms like Youtube.

IIT Bombay | Research Assistant (AISTATS 2026) | Mumbai, India

Jul'23 - Jan'25

- Designed the novel **Selective Policy Improvement (SPI)** algorithm to compute optimal policies for **MDPs with state sensing costs**, outperforming state-of-the-art Partially Observable MDP solvers across all sensing costs.

Google | Silicon Engineering Intern | Bangalore, India

May'23 - Jul'23

- Optimized **design verification** process by **15%** with toggle coverage analysis using Python-based automation
- Developed **automated checkers** for data retention flops in low-power mode applications, streamlining verification

Selected ML Projects

Imitation Learning from Denoising Diffusion Models

Sep'24 - Nov'24

- Developed **transformer-based diffusion policies** for imitation learning in BipedalWalker OpenAI Gym environment, achieving **98% of expert performance** and benchmarking against algorithms like DAgger and Behavior Cloning

Text-to-Image Diffusion Models with Enhanced Semantic Understanding

Jan'24 - May'24

- Devised SUR (Semantic Understanding & Reasoning) architecture utilizing **LLAMA-based prompt enrichment**
- Boosted **multi-modal** visual question answering accuracy (counting/color/action) by **20%** over baseline CLIP

ToyGPT: Miniature GPT for Arithmetic Addition

Sep'24 - Nov'24

- Trained small GPTs for 2-digit addition task evaluating **Positional Encoding variants** and Layer-Norm architectures

RL in Billiards and Football Half-field Offense

Jul'23 - Dec'23

- Implemented **Monte-Carlo Tree Search (MCTS)** for potting balls in minimal attempts (< 10) for noisy billiards
- Executed MDP Planning to devise an optimal half-field football offense strategy using value and policy iteration

Skills

Programming Languages:

Python (Proficient), C++, Julia, Java, MATLAB, HTML

Libraries:

PyTorch, JAX, TensorFlow, Keras, Hugging Face, PyTorch-Geometric, Scikit-learn, Pandas, SciPy, SymPy, OpenCV, NLTK